700105 Simulation & Concurrency

UNIVERSITY of HULL

Simulation & Concurrency

- Simon Grey
  - S.Grey@hull.ac.uk
- Warren Viant
  - W.J.Viant@hull.ac.uk

Teaching Schedule

- Lecture/Workshops
  - Tuesdays 9:00-11:00 RBB-335
  - Thursdays 9:00-11:00 RBB-335
- Labs
  - Thursdays 13:00-15:00 RBB-335

Competencies

- Students will be able independently to Identify and apply appropriate logic and mathematical approaches to computational problems, demonstrating computational thinking in designing, developing, testing and critically evaluating systems.
- Students will be able independently to select, design, implement and critically evaluate a range of algorithms according to functional and non-functional requirements.
- Students will be able independently to demonstrate game programming skills by developing programming solutions to significant problems using a range of technologies, with an emphasis on working towards finished, potentially shippable artefacts.

Developing a Library

- Students will be able independently to demonstrate game programming skills by developing programming solutions to significant problems using a range of technologies, with an emphasis on working towards finished, potentially shippable artefacts.
- “potentially shippable artefacts” does not mean “finished game”
- In this module we will attempt to build our own physics engine

CoPilot/Chat GPT

- If you ask the right questions CoPilot will do most of the work for you
- Here the focus is to help you understand the right questions to ask, and try to ensure that you understand the answer
- Part of that can be done through testing for specific scenarios and edge cases

Assessment

- Series of lab small exercises throughout
  - You will demonstrate these during the lab sessions
- Single large lab task bringing everything together
  - This is where the majority of the marks come from

Simulation

Roadmap

- Introduce the topic
- Create a library
- Create a test project to test the library using unit tests
- Create a framework to load scenes and render them

Definitions – Oxford English Dictionary

- imitation of a situation or process.
  - "simulation of blood flowing through arteries and veins"
- the action of pretending; deception.
  - "clever simulation that's good enough to trick you"
- the production of a computer model of something, especially for the purpose of study.
  - "the method was tested by computer simulation"

Definitions

- **Simulation** *"A computer model of a real phenomenon or system. A 3D simulation is described by 3D models in a computer program. Simulations are used in computer games, training programs (flight simulators) and by scientists, who recreate, project into the future and predict real world phenomena."* 3dgraphics.about.com
- **Physically-Based Modelling** *"Physically Based Modelling is modelling that incorporates physical characteristics into modelling allowing numerical simulation of their behaviour"* Ronen Barzel

Definitions takeaway

- Simulations are ***models*** and all models have limitations, cut corners and take shortcuts.
- We need to understand the benefits and implications.

Simulation in Games

- Updating state
- Figuring out if things hit each other
- Figuring out what you should do about it
- Ways to reduce calculations

Simulations in Games

- Knowing what corners you can cut
  - What level of detail do you need?
  - Understanding how the virtual world is represented
  - What data is dynamic and needs to be computed each frame?
  - What data is static?
  - What could be generated as part of the development process?

Indicative Content

- Updating State
  - Calculating forces
    - Gravity, Drag, Friction
  - Calculating new velocities and positions
    - Integration methods (Euler, Semi-Implicit Euler, Verlet, Runge-Kutta)
- Collisions - Geometric Objects (Shapes)
  - Primitives (point, line, line segment, plane, sphere, cylinder, capsule, cuboid, triangle, convex hull)
  - Methods (Inside-Outside, Intersects, Contains, SAT, GJK)
- Collision Response
  - Calculating new velocities or forces after collision
  - Conservation of momentum
- Reducing Calculations
  - Spatial Data Structures (uniform grid, quadtree, octree, kd-tree, BSP tree)

Approach

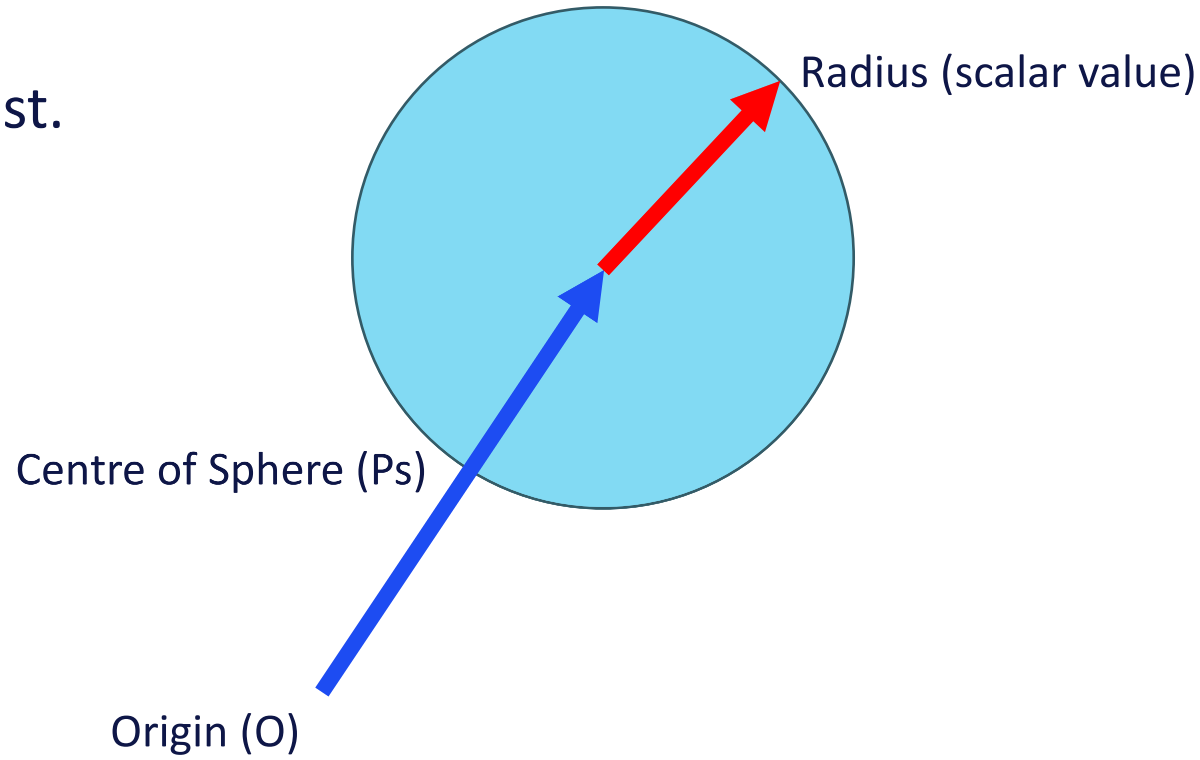
- Independently develop a selective maths library, supported with Unit tests
  - We will start this today
- Independently develop a greybox/sandbox environment that you can load simulation scenarios into that use the maths library
  - Use C++, Vulkan and ImGui

Getting Testing Set Up

- I’m suggesting using GoogleTest
  - GoogleTest User’s Guide | GoogleTest
- Other testing frameworks are available

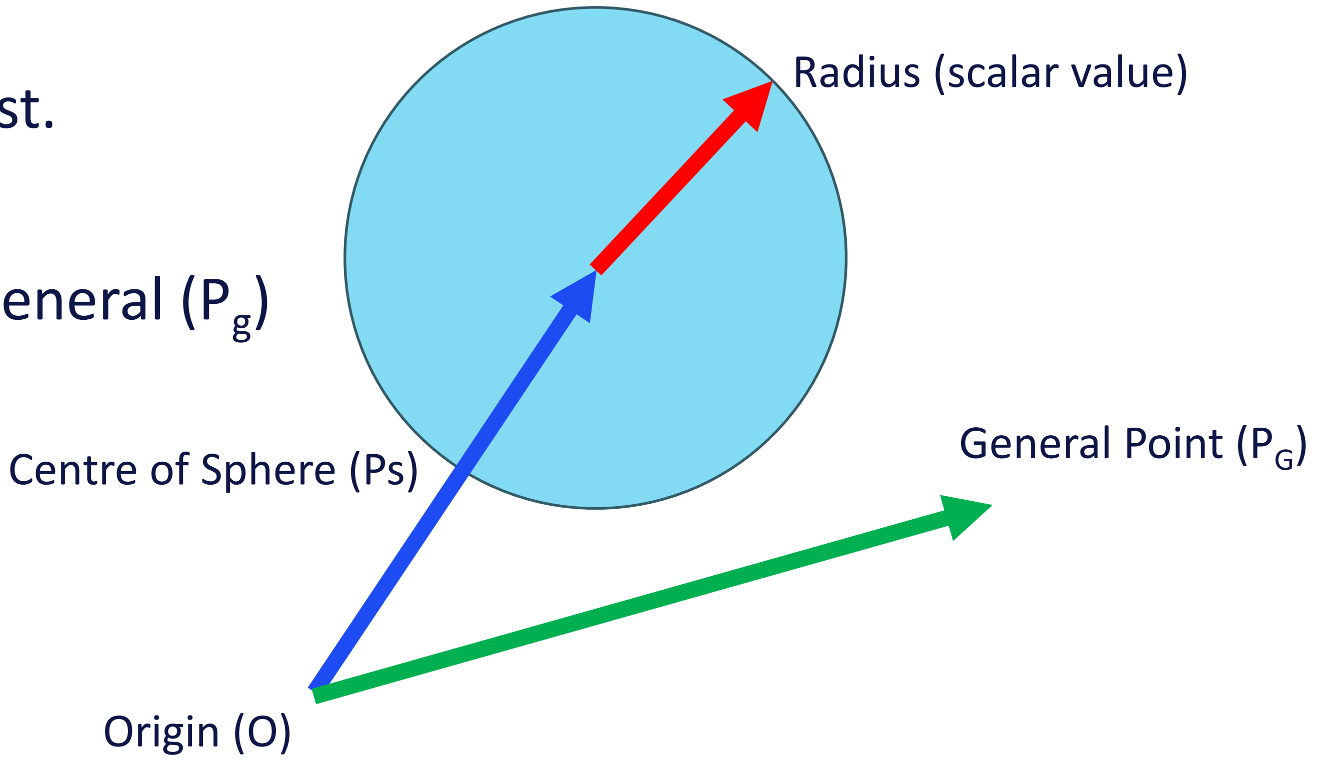
Sphere

- We need something quick to test.
- We're going to make a Sphere



Sphere

- We need something quick to test.
- We're going to make a Sphere
- Then we're going to check if a general (P_g) point IsInside the sphere

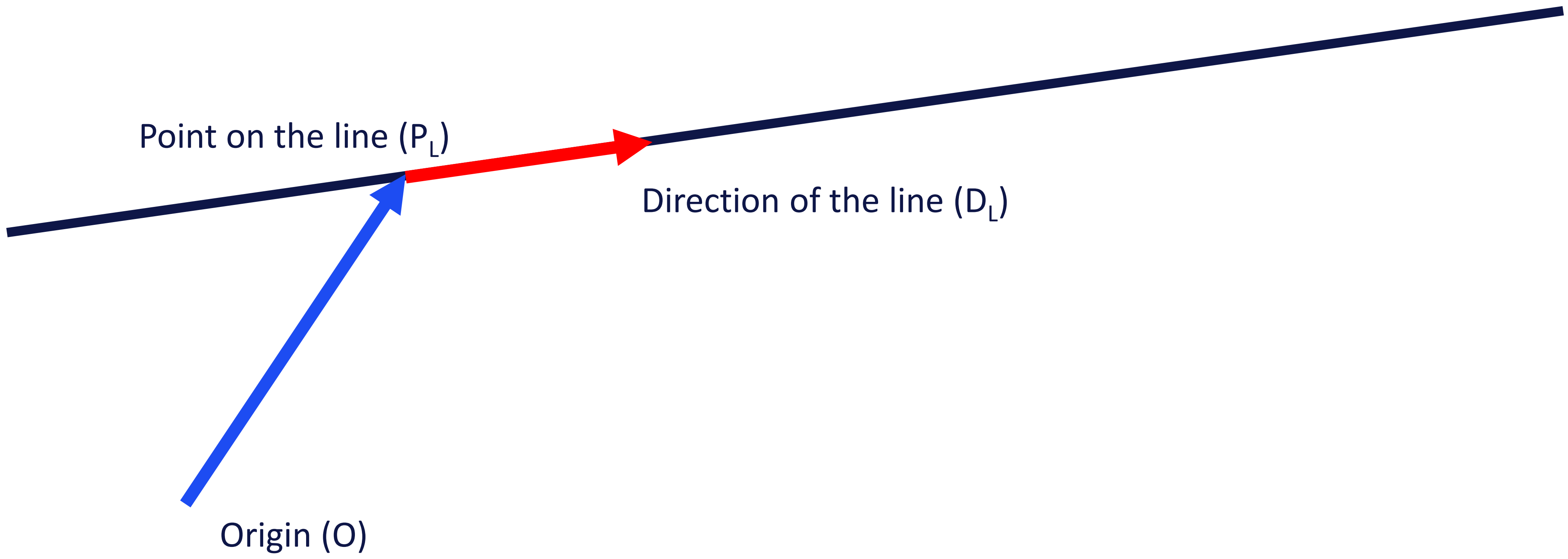


Time to Code

Canvas Workshop 1.1 Setting Up Testing

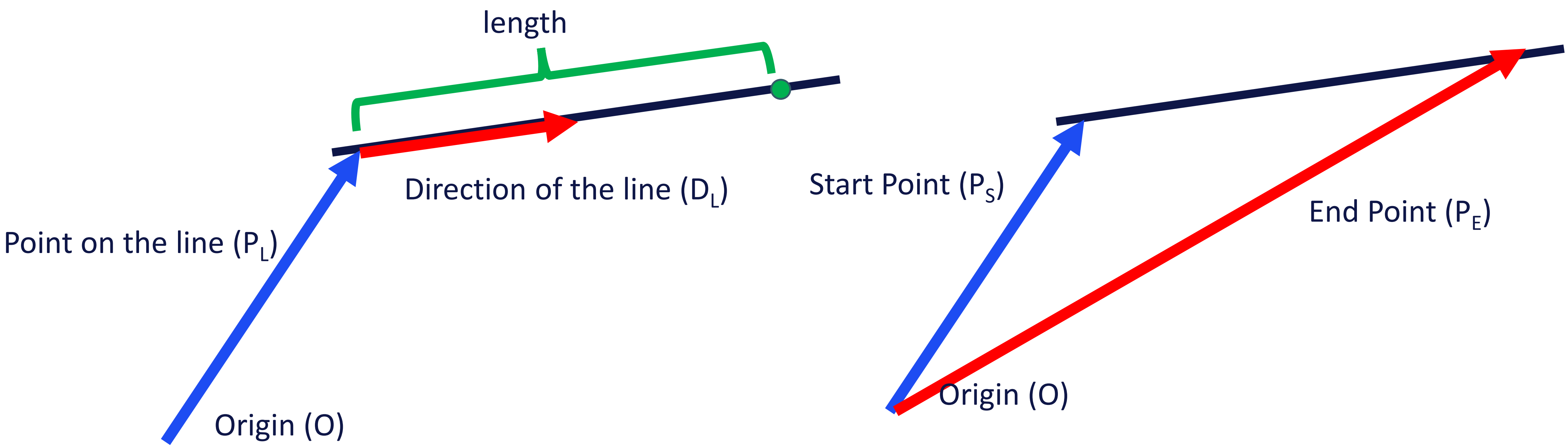
Lines/Rays

- Glm provides you with Vector functionality that you should be familiar with
- Points on the line can be found using $P_L + mD_L$
- D_L is ***typically*** unit



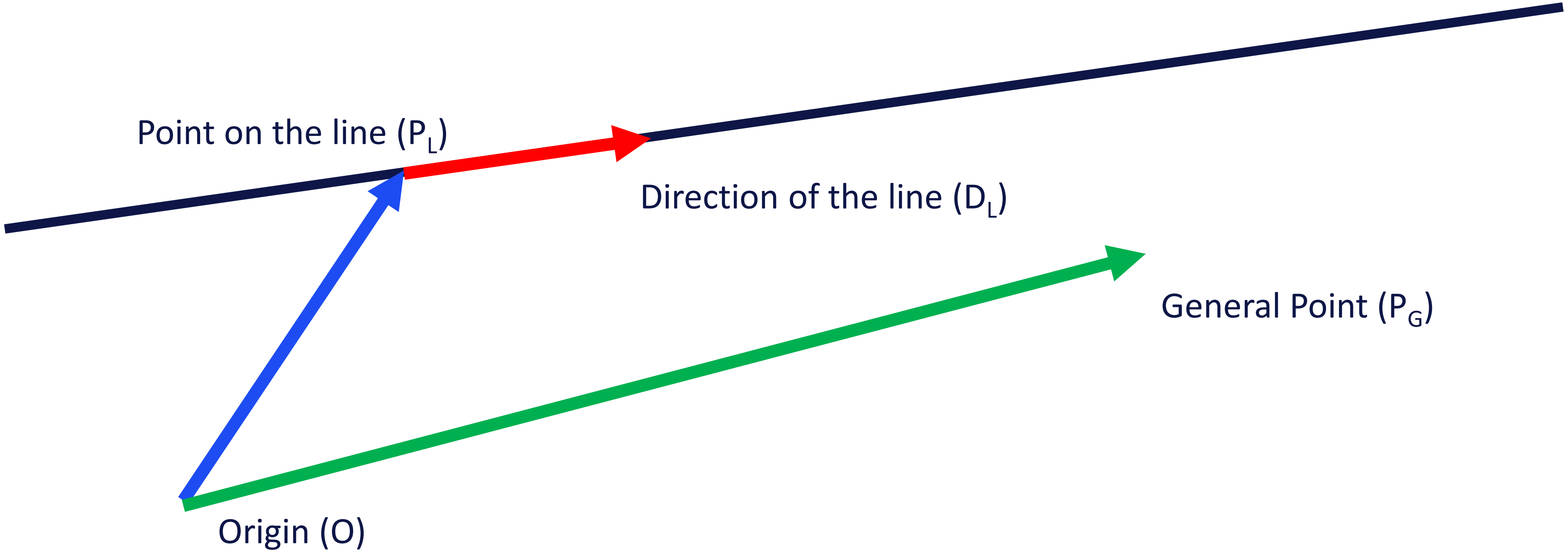
Line Segments

- A Line with a Length
- Could be represented as a Point P_L, a Direction D_L and a length
  - Could be represented as an instance of Line and a length
- Could be represented as the start point and end point



Shortest Distance From a Point to a Line

- Points on line are $P_L + mD_L$
- Remember P_L and D_L are 3D vectors with x, y and z components



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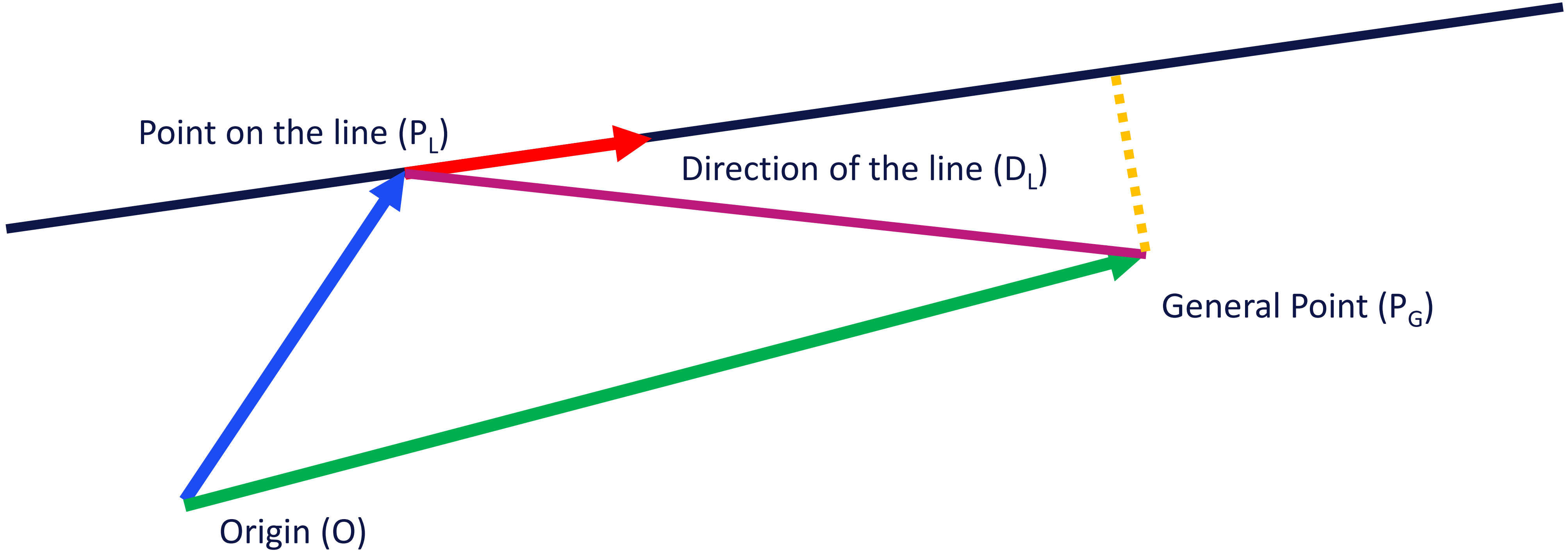

Additional Resources

- 2D solutions for same problem
- https://en.wikipedia.org/wiki/Distance_from_a_point_to_a_line#:~:text=The%20distance%20(or%20perpendicular%20distance,is%20perpendicular%20to%20the%20line
  .

Time to Code

Canvas Workshop 1.2 Testing Sphere-Line Intersection and Sphere-LineSegment Intersection